

Machine Learning Ethics Game

Debriefing Session

April 28, 2026

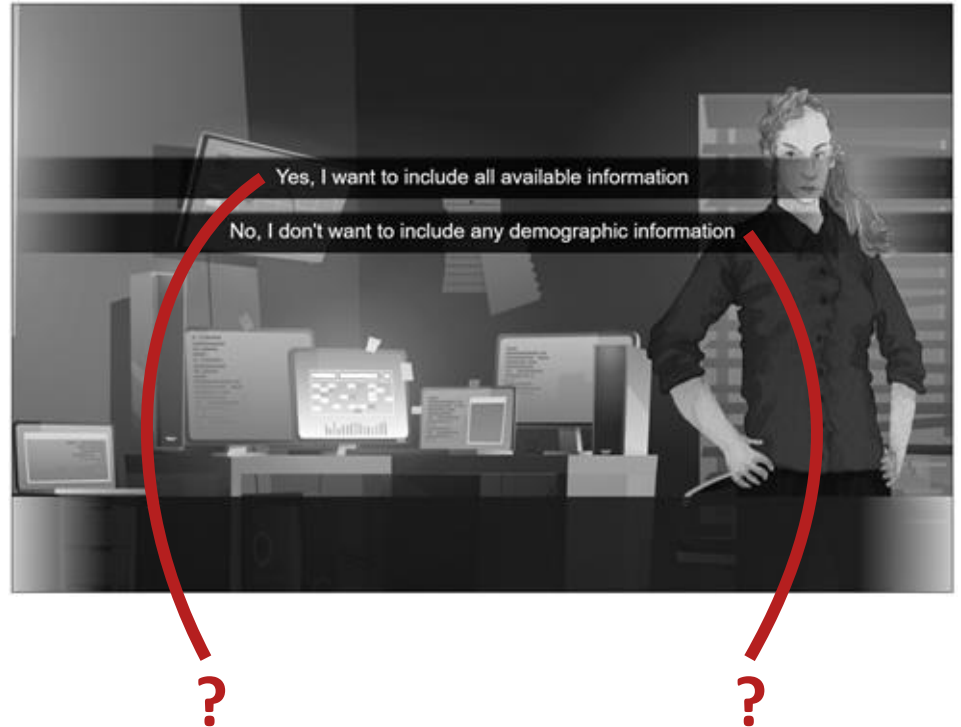
Let's refresh our memory

In the prisoner case:

- What decision did you take: include the demographic information or not ?
- What was the consequence?

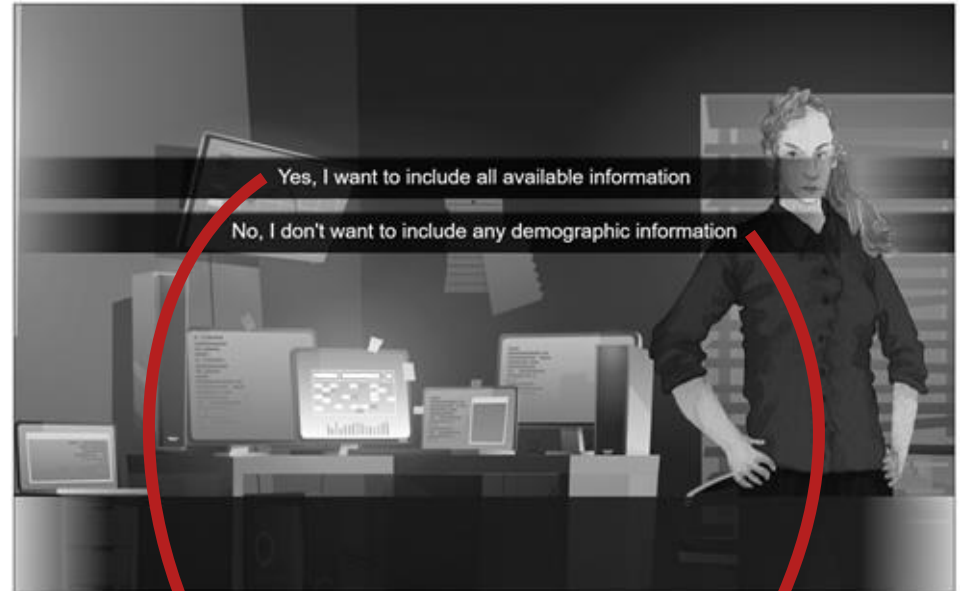
Discuss with your neighbor:

- What are the consequences for each of these choices?



Accuracy vs. fairness

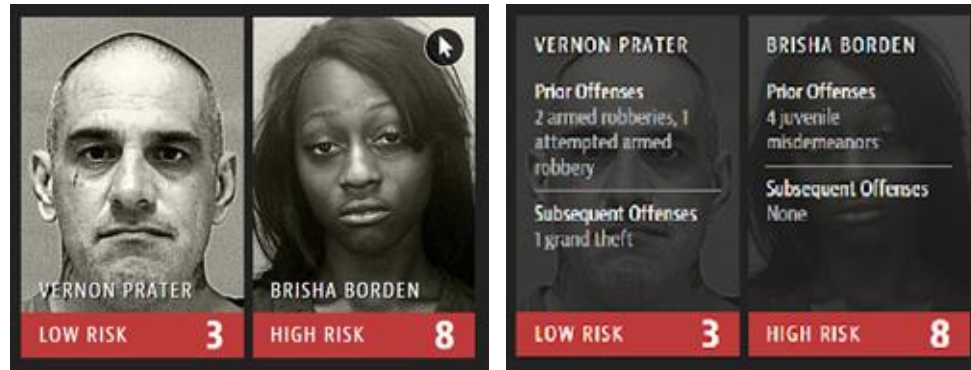
Pessach, D., & Shmueli, E. (2022). A Review on Fairness in Machine Learning. ACM Computing Surveys, 55(3), 51:1-51:44.
<https://doi.org/10.1145/3494672>



Racist model
► **Friend not released**

Low accuracy
► **Neighbor robbed**

Based on real cases



Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016, May 23). Machine Bias : There's software used across the country to predict future criminals. And it's biased against blacks. ProPublica.

<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

Dressel, J., & Farid, H. (2018). The accuracy, fairness, and limits of predicting recidivism. Science Advances, 4(1), eaao5580.

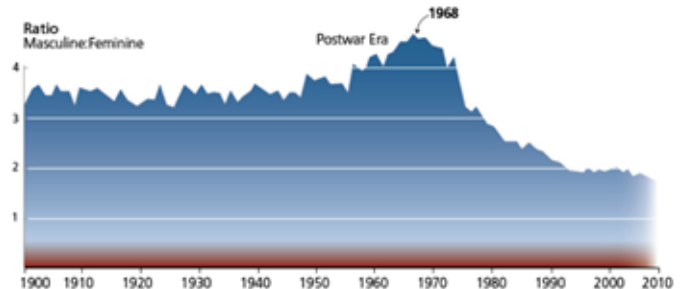
<https://doi.org/10.1126/sciadv.aao5580>



Machine Translation | Gendered Innovations

<http://genderedinnovations.stanford.edu/case-studies/nlp.html>

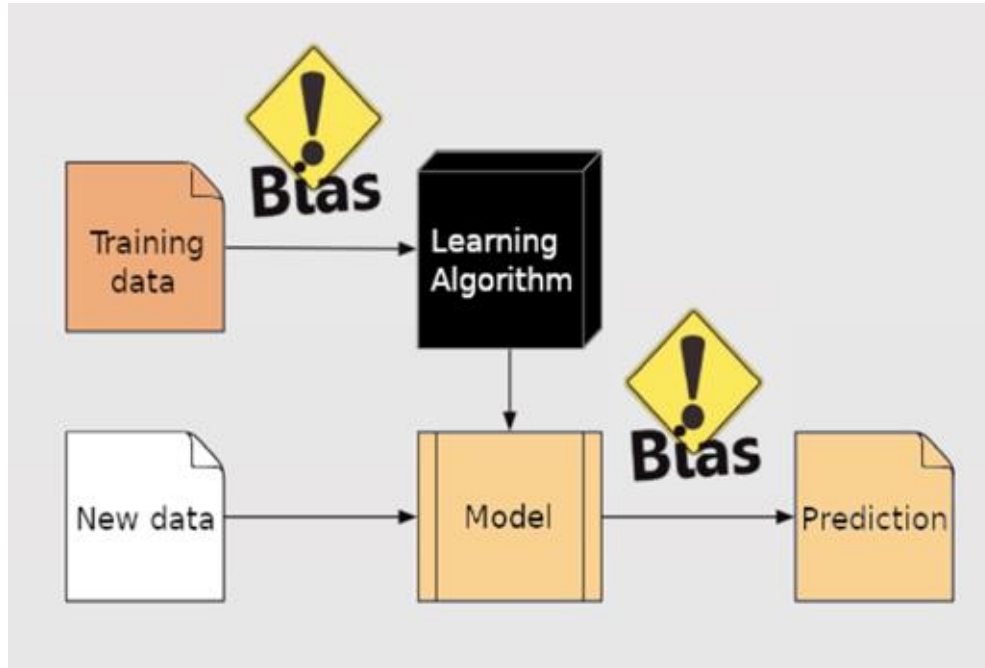
Ratio of Masculine to Feminine Pronouns in U.S. Books, 1900-2008
Changes parallel increases in women's labor force participation, education, age at first marriage, etc.



The ratio of masculine pronouns ("he," "him," "his," "himself") to feminine pronouns ("she," "her," "hers," "herself") peaked at over 4:1 in 1968. By 2000 the ratio dropped dramatically to 2:1 (Twenge et al., 2012).

Data from American English corpus of the Google Books database (~1.2 million books).
Reproduced from Twenge et al., 2012.

Bias and fairness in Machine Learning



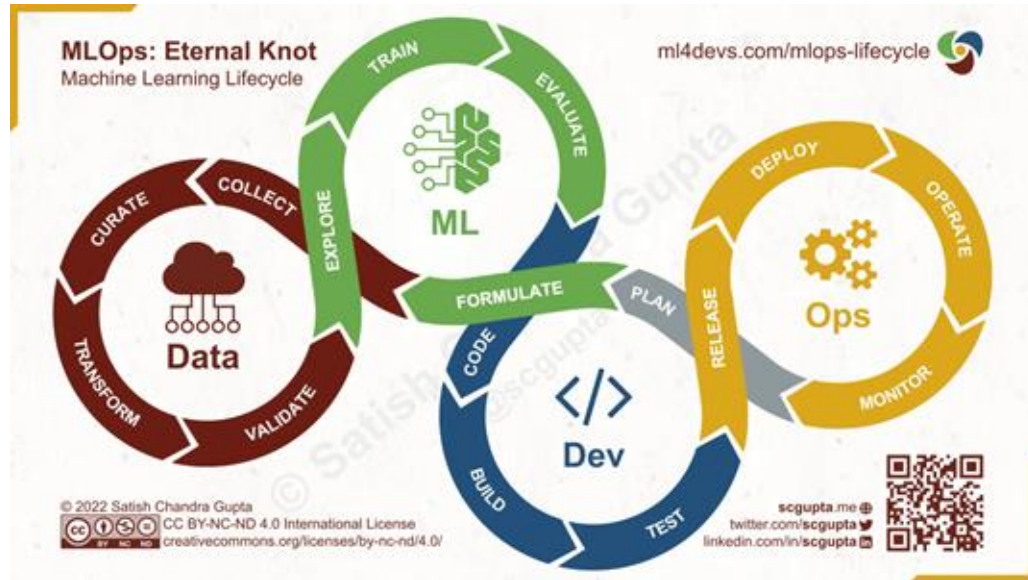
Most Machine Learning systems will affect **people**

“Crowd” by Amy West on Flickr, CC BY 2.0

https://www.flickr.com/photos/amy_elizabeth_west/3876549126/



Your **decisions** can make a difference!

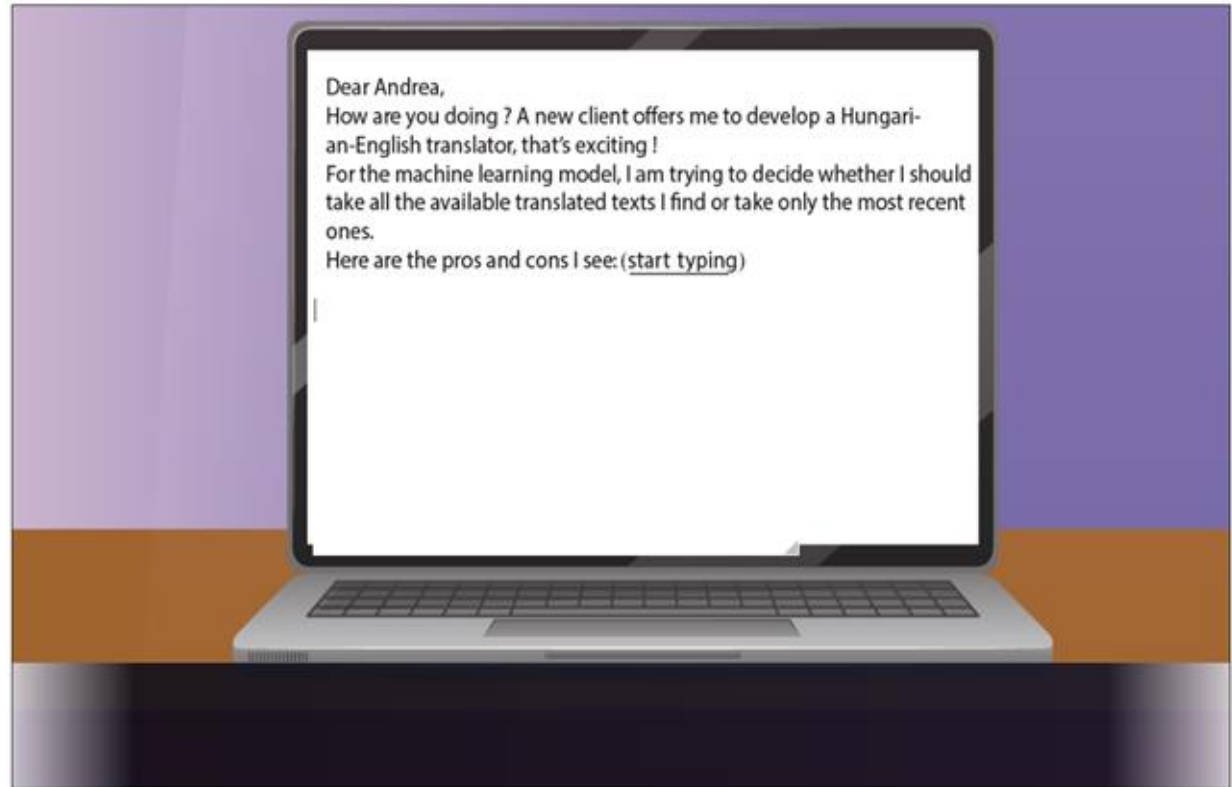


“It is important to acknowledge that **not all problems should be blamed on the data**. The ML pipeline involves a series of choices and practices, from model definition to user interfaces used upon deployment. **Each stage involves decisions that can lead to undesirable effects.**”

(Suresh & Guttag, 2021)

Responsible design decisions

→ Slow down
your decision
process



Responsible design decisions

→ Think about who could be affected and how?

- Individuals / communities?
 - Majority / minorities?
 - Vulnerable populations?
 - Nations / society?
 - Companies / staff?
 - ...
- Benefit / harm?
 - High / low risk?
 - Direct / indirect impact?
 - Short / mid / long term?
 - Privacy / security / human rights?
 - Misuse, attacks?
 - ...

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Work with a
diverse team

Create
speculative
scenarios

Carry out
research

Involve the
publics

Example case: a ML model predicts user emotions based on smartphone touch data

Imagine where this system could be used (e.g. messaging apps, social media) and **what could go wrong** for:

- You and your family or friends?
 - Children? Teens?
 - People in a dictatorial country?
 - Workers in companies?
 - What if it's misused? Attacked?
-
- Post **1 message per idea**